

3(a)	upthrust, weight	B1
3(b)	upthrust greater than weight so (resultant force is) upwards	B1
3(c)(i)	A, g and ρ all constant so $F \propto x$	B1
	minus sign means F and x are in opposite directions	B1
3(c)(ii)	$(a = \frac{F}{m} \text{ so } a = (-)\frac{Ag\rho x}{m})$	M1
	so $\omega^2 = \frac{Ag\rho}{m}$ hence $\omega = \sqrt{\frac{Ag\rho}{m}}$	A1
3(d)(i)	damping due to viscous forces	B1
3(d)(ii)	$(E =) \frac{1}{2}m\omega^2 x_0^2$	C1
	$\omega^2 = (-)$ gradient	C1
	$(E =) \frac{1}{2}m\omega^2 (x_1^2 - x_2^2)$ $= \frac{1}{2} \times 0.57 \times (2.3/0.020)(0.020^2 - 0.016^2)$ $= 4.7 \times 10^{-3} \text{ J}$	A1